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YOGA AND CANCER

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ABSTRACT

The research compiled here suggest that yoga could aid at different phases of the disease and its treatment to enhance quality of life, lessen fatigue, boost activity and fitness levels, improve sleep quality, increase appetite, and lessen worry. yoga, meditation, may be able to benefit increase tolerance to pain, endure side effects of cancer therapy and improve emotional scores in the patients. The plausible evidence based mechanisms involved have been reviewed here.

INTRODUCTION: HYPOXIA AND CANCER

A study of 160 patients with breast cancer who were undergoing radiation therapy were randomized to pranayama or control group and their emotional aspects such as impatience, tiredness, worry, anxiety and frustration were assessed (Chakrabarty et al., 2016). Pranayama included bhramari, sheetali and nadishodhana for 20-30 minutes in morning and evening 6 days per week for 6 weeks. Pranayama group resulted in about 70 percent reduction in these negative emotions than the control group. Chronic hypoxia is linked to cardiovascular disease, cancer, and neurodegenerative conditions by hypoxia inducible factor-1 (HIF-1), a transcription factor meditating changes in gene expression essential to meet the needs of reduced O2 (Michiels, 2004). In the short-term HIF-1 supports adaptive strategies for managing diminishing SAO2, but when persistently elevated, HIF-1 triggers pathways that generate oxidative stress and mitochondrial damage (Thomas & Ashcroft, 2019). Chronically elevated HIF-1 communicates with another transcription factor, nuclear factor of the kappa-light-chain-enhancer of activated B cells (NF-KB), notorious for upregulating genes associated with disease and inflammation (Acker, 2004). Additionally, when SAO2 is lower, vagal output to the heart is withdrawn, resulting in a higher resting heart rate (Siebenmann et al., 2019). Although temporarily useful to hasten oxygen delivery, sustained elevations of resting heart rate and autonomic dysregulation correlate with reduced physiological and psychological resiliency. Consequently, even marginally reduced SAO2 can be pernicious. Practicing VRBPs that increase SAO2 may help prevent or reverse these pathogenic changes.

ROLE OF PRANAYAMA IN CANCER CARE

William G. Kaelin Jr., MD, won the 2019 Nobel Prize in Physiology or Medicine with two other physician-scientists for unraveling a molecular mechanism that not only is crucial to survival, but is entwined with cancer and other diseases. He showed that cancer basically needs a low oxygen environment to survive. He suggested that cancer cells "live in hypoxic, very low oxygen, and acidic conditions and derive energy from sugars by fermenting them the way yeast does." From this, he theorized that these low-oxygen and highly-acidic conditions caused cancer. Recently, scientists at The Johns Hopkins University, based on their experiments conducted on human breast cancer cells and mice, explain how certain cancer stem cells thrive in low oxygen conditions (Johns Hopkins Medicine, 2016). Thus, it is the author's conjecture that when the body cells have sufficient oxygen, cancer may be prevented or at least delayed. Raa and colleague (2007) have reported that tumor growth was significantly reduced after hyperbaric oxygen treatment compared to control and even more than chemotherapy in mammary tumors inducted in rats.

Recent review by Danhauer and colleague (2017) of studies on effect of yoga interventions including breathing exercises on cancer patients during treatment by chemotherapy concludes that there is evidence to recommend yoga for improving psychological outcomes (such as reduction of anxiety, depression and distress), with potential for also improving physical symptoms.

Breast cancer patients showed an improvement in physical functioning based on the physical component scale scores after a 6-week yoga program. The quality of life and stress symptoms showed greater improvements in cancer affected patients when subjected to mindfulness meditation and gentle yoga (Woodyard, 2011).

LYMPHATIC AND IMMUNE HEALTH

Cancer patients experience stress-related changes in immunity, which may include decreases in cell mediated immunity as well as increases in inflammatory cytokines (Lutgendorf & Sood, 2011). Indeed, chronic inflammation is a major risk factor for cancer, and also associated with poor outcomes in cancer patients (Wu et al., 2014). A systematic review of 15 RCTs concluded that yoga may improve immune system function by the down regulation of pro-inflammatory markers (including IL-1beta, IL-6, and TNFalpha), as well as by the enhancement of cell-mediated and mucosal immunity, although small sample sizes and heterogeneity in study design precluded performing a meta-analysis (Falkenberg et al., 2018). Cancer patients have to deal with the direct consequences of unregulated cell growth in the organ or tissue affected, the toxic consequences of their treatments, as well as the psychosocial effects of their diagnosis. Cancer related fatigue (CRF) is a serious problem which interferes with the quality of life of cancer patients (Abrahams et al., 2016). A meta-analysis of 2183 subjects diagnosed with breast cancer enrolled in 17 RCTs assigned to either a yoga intervention or a varied control group found that yoga had a large impact on the mitigation of physical fatigue, a medium impact on cognitive fatigue, and a small impact on mental fatigue in the breast cancer patients (Dong et al., 2019). Another meta-analysis of 245 studies found that yoga as well as a number of other exercise and non-pharmacological interventions had moderate to large effects on the mitigation of CRF in cancer patients with a variety of different cancer types while they were undergoing treatment, and that yoga had the largest effect of all the modalities evaluated post treatment (Hilfiker et al., 2018). A meta-analysis of 41 RCTs evaluated the efficacy of a number of non-pharmacological interventions including standard exercise to mitigate depressive symptoms in breast cancer patients relative to standard care. Yoga, mindfulness and psychotherapy emerged as the interventions that significantly reduced depressive symptoms. Systematic reviews have reported that yoga may improve depression, distress, anxiety, sleep quality, fatigue, musculoskeletal symptoms, as well as biomarkers of stress, inflammation, and immune function in cancer patients (Danhauer et al., 2019; Danhauer et al., 2017; Lin et al., 2018). A comprehensive review of 138 clinical trials in which 10,660 cancer patients from 20 countries were studied found strong support for the integration of yoga into standard care for cancer patients and noted that while studies on the role of yoga in cancer prevention are lacking, this is an area that is ripe for future investigation (Agarwal & Maroko-Afek, 2018).

CELLULAR HEALTH

Telomeres are short sequences of DNA that protect the ends of chromosomes. Telomeres shorten with each division of a somatic cell, and as such are correlated with cellular aging. Cells with shortened telomeres are at risk of apoptosis (cell death), and shortened telomeres have been associated with atherosclerosis, diabetes, chronic stress and depression (Zhang et al., 2014). However, telomeres can also lengthen in a reaction catalyzed by the enzyme telomerase (Epel et al., 2009). A systematic review was conducted of 12 RCTs in which the effects of yoga practices on telomere length and telomerase activity were studied (Rathore & Abraham, 2018). Three studies looked at radiation induced telomere shortening in breast cancer patients; two of the three studies found less telomere shortening in the yoga group versus the control. Five additional studies measured telomere length in varied populations; four of the five studies found telomeres were longer in the mindfulness practitioners relative to controls. Six studies measured telomerase; five of the six studies found it to be higher at all time points measured in the mindfulness practitioners relative to controls. While these studies are small, they constitute emerging evidence that yoga practices including asana, pranayama, and meditation maintain telomere length. The mechanisms are suggested to include increased delivery of oxygen to cells through yoga practices such as asana and pranayama, as well as stress reduction through regulation of the HPA axis by a wide range of mindfulness practices (Epel et al., 2009; Rathore & Abraham, 2018).

It is known that overweight and obesity are associated with an increased risk of the diseases (viz. cardiovascular problems, type 2 diabetes mellitus, stroke, asthma, arthritis and certain cancers) (Telles et al., 2018). Apart from this, obesity is also known to impair the several aspects of quality of life (Telles et al., 2019). Overweight and obesity places an enormous burden on economy through (i) direct cost (e.g., spending on diagnosis and treatment of medical conditions associated with obesity) and (ii) indirect costs (e.g., being overweight/or obese reduces productivity and negatively influences physical and mental well-being) (Dee et al., 2014). It is estimated that relative medical expenses of obese adults are 100 times higher than normal weight adults. Hence it is apparent that if the prevalence of obesity continues to rise, it will place a huge burden on health care sector as well as on economy. Lifestyle interventions consisting of physical activity, dietary modifications and changes in thinking or perception are considered effective for the prevention and management of obesity (Galani & Schneider, 2007). Lifestyle interventions of different durations (8-24 weeks) have been reported to reduce 5-10 percent of initial body weight (Jensen et al., 2014). These changes in body weight through lifestyle interventions are considered of clinical importance as these changes in initially body weight are associated with a decrease in metabolic and cardiovascular risk factors which have been linked with obesity (Jensen et al., 2014). However it has been reported that overweight and obese persons experience physical and psychological challenges, due to compromised physical and psychological health, which make it difficult for them to adhere to a lifestyle intervention that includes an increased level of physical activity and dietary modifications. Hence tailored behavioral interventions which are safe, yet effective, are recommended for overweight and obese persons. Yoga is a comprehensive lifestyle intervention which includes an increased level of physical activity, dietary modifications, guided relaxation and changes in thinking and perception (Telles et al., 2014). The popularity of yoga, as a weight loss intervention, is increasing rapidly. The increase in popularity is primarily due to the fact that the practices of yoga can be modified according to the need of the persons with varied conditions associated with obesity; hence the practices are considered relatively safe for overweight and obese persons. With this background the practice of yoga has been assessed for its effectiveness in the management of obesity with studies suggesting long term adherence (Yu et al., 2018). The aims of the present review were to (i) review the studies assessing the effects of yoga practices on obesity and (ii) grade them according to the level of evidence using standard grading method.

Breast cancer survivors who were obese were assessed in a randomized controlled trial (Littman et al., 2012). The study aimed at evaluating yoga in this group with special emphasis on its efficacy on fatigue, quality of life and change in weight. The participants were 63 post treatment breast cancer survivors with ages between 21 and 75 years; a wide range which would include pre and post menopausal women, which is not ideal. Their BMI was ≥ 24 kg/m2 (or ≥ 23 kg/m2 if they were of Asian descent). The sample size was calculated prior to the trial and n = 60 was considered adequate to demonstrate any changes. Participants were block randomized to a yoga or to a wait-list control group, based on (i) age (strata 21- 49 years, 50-69 years, 70-75 years), (ii) stage (2 strata: 0/I grade and II/III grade) and BMI (2 strata: 24-29.9 kg/m2 and ≥ 30 kg/m2). This detailed stratified sampling helped to ensure comparability of the two groups. After stratification participants were randomized to the two interventions. The variables assessed were intended to understand the effect of the intervention on cancer as well as on obesity; and included Functional Assessment of Cancer Therapy General (FACT-G), a quality of life measure (QOL); physical activity, anthropometric measures, class attendance, home practice, knowledge of yoga and feasibility (based on time for recruitment, frequency of attendance, and participant's satisfaction, among other factors). Questionnaires were not blind scored. Baseline data were compared with Chi-Square and t-tests. An intent-to-treat approach was used. Linear regression was used to assess changes in outcomes between groups, adjusted for baseline values. The yoga intervention was based on Vini yoga a Hatha yoga type of therapeutic yoga which includes physical stretches, postures, breath control and meditation. Yoga was to be practiced at least 5 times in a week with one 75 minutes class at the facility. There was no dietary advice. The wait list control group was asked not to practice yoga. After the 6 months follow up assessments they were given the option and facilities to practice yoga. The study was a stratified randomized controlled trial, with assessments at baseline and after 6 months. Adverse events were looked for, but none were reported. The yoga group showed better quality of life, reduced fatigue, and a decrease in waist circumference. However none of the findings were statistically significant. It is a limitation of the study that the sample size was not adequate for potentially clinically important effects to achieve statistical significance. Other limitations were that the control group did not receive additional attention which the yoga group received, hence it is difficult to understand the exact contribution of yoga practice per se, and of additional attention along with yoga. The lack of blind scoring and of objective quantitative measures are other limitations of the study. Though this study was partially conducted in a facility for cancer treatment, with the remaining sessions as home practice, there have been studies which have tried residential yoga programs in India (Telles, et al., 2010).

Shoulder stretching is essential to improve breast cancer-related lymphedema, kastatarkshna asana, rajjukarshana helps in shoulder stretching, thus improves the shoulder strength.

Just one session of relaxing practices like meditation, yoga, and chanting influenced the expression of genes in both short-term and long-term practitioners, according to a Harvard study. Blood samples taken before and after the breathing practices indicated a post-practice increase in genetic material involved in improving metabolism and a suppression of genetic pathways linked with inflammation. Since chronic inflammation has also been associated with such deadly diseases as Alzheimer's, depression, cancer, and heart disease, it's probably fair to say that better breathing may not only change your life but may also save it.

Two types of happiness are described by Aristotle- eudemonic and hedonic. Eudemonic happiness stems from this realizing of one's potential and is described as a non-transitory, steadfast contentment or joy. Hedonic happiness is considered a more transitory emotional experience, such as cheerfulness (Aristotle et al., 2004). The significance of eudemonic well-being is demonstrated by the growing body of research that supports its widespread and positive health effects for various patient populations. These benefits include: 1. Lower levels of fatigue, disability, pain intensity, pain medication use, depressive symptoms, and improved patient functioning and life satisfaction in people with chronic pain conditions (Dezutter et al., 2013; Dezutter et al., 2015; Schleicher et al., 2005). 2. Decreased all-cause mortality independent of factors such as age, physical inactivity, conditions such as cardiovascular disease or cancers (Keyes & Simoes, 2012; Pressman et al., 2019). 3. Eudemonic well-being is differentiated from hedonic wellbeing with its effect on inflammatory and immune processes. While eudemonic well-being is associated with a down-regulation of gene profiles related to healthier inflammatory and immune processes affecting neurodegenerative, cardiovascular conditions and cancers- hedonic well-being is associated with its up-regulation (Cole et al., 2015; Fredrickson et al., 2013; Fredrickson et al., 2015). 4. Eudemonic well-being is correlated with decreased levels of salivary cortisol, pro-inflammatory cytokines, and lessened cardiovascular disease risk, hedonic well-being has minimal effect on these biomarkers (Ryff et al., 2004). These myriad and distinct health benefits demonstrate eudemonic well-being as an important contributor to physical, psychological, and social health.

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